



ADLynk

Multi-Channel Advertising Management Platform

Database Systems — Project Proposal

Course Database Systems

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1 Introduction

Advertising is a key part of modern business, but managing campaigns across multiple platforms can be time-consuming and complex. Businesses often rely on different agencies and services, which makes coordination difficult. To address this, we propose a Multi-Channel Advertising Management Platform that consolidates all advertising options into a single digital system.

The platform enables business owners to manage campaigns from one dashboard, allowing them to book services such as online promotions, social media marketing, billboard advertising, and vehicle branding. Business owners can also reserve billboard spaces in various cities across Pakistan through an organized interface.

Using a centralized database, the system efficiently stores and manages information on users, advertising services, billboard locations, bookings and payments. This ensures streamlined data management, transparent pricing, and easy campaign tracking.

Developed under ADLink, the platform makes professional marketing accessible to large businesses, startups, and small online store owners, simplifying the advertising process and improving operational efficiency.

2 Problem Statement

The current advertising system in Pakistan has several major problems that make it difficult to manage campaigns and advertise businesses effectively.

- **Lack of Integration:** : Business owners have to deal with billboard owners, vehicle branding companies and advertisers separately, usually through informal communication, because there is no single platform that connects them all.
- **Lack of Transparency:** There is no proper centralized system to find advertising services, check prices, see availability, or confirm bookings, which makes the advertising process hectic and complex.
- **Manual Processes:** Campaign tracking, payments, and bookings are often done manually using spreadsheets or phone calls, which keeps the data separate and hard to manage.
- **No Centralized Booking:** Without a structured booking system, double-bookings, cancellations, and unresolved disputes are common, damaging business relationships.

Overall, Currently there is no unified system that connects businesses with advertising service providers. This makes businesses spend more time managing these complex processes instead of focusing on improving their products, services, and growth.

ADLYNK DIRECTLY SOLVES THESE PROBLEMS

ADLink provides a centralized platform that brings billboard owners, advertisers, business owners and vehicle branding companies, together in one system. Businesses can easily discover available advertising services, view prices, and check details, while bookings

and payments are managed securely within the platform. All campaigns, services, and transactions are stored in an organized database, reducing manual work and inefficiencies. This streamlined approach allows businesses to focus on improving their products, services, and growth, rather than dealing with the complexities of managing advertising campaigns.

3 Objectives

The main objectives of the ADLink Multi-Channel Advertising Management Platform are:

1. To **integrate all advertising services** in one platform, so business owners no longer need to contact billboard owners, vehicle branding companies, and advertisers separately.
2. To provide **transparency in service discovery**, pricing, availability, and booking confirmation, reducing a hectic and complex advertising process.
3. To allow businesses to **book and manage campaigns** directly through the platform, reducing manual effort and complexities.
4. To **automate campaign tracking, payments, and bookings**, eliminating manual spreadsheets and phone call processes.
5. To establish a **centralized booking system** that prevents double-bookings, cancellations, and unresolved disputes.
6. To allow businesses to **focus on improving products, services, and scaling**, rather than managing advertising complexities.
7. To provide a centralized platform where businesses—including small e-commerce stores—can easily access high-quality advertising services, manage campaigns, and make secure bookings, while service providers can reach clients efficiently, ensuring better services and smooth operations for everyone.

4 Scope

The ADLink Multi-Channel Advertising Management Platform is designed to provide a centralized, efficient, and transparent system for managing advertising services. The scope of the system includes the following key features:

Included Features:

- All users—including business owners, service providers, asset owners and admins—can register and log in securely. The system provides role-based access so that each user can access only the features relevant to their role.
- Service providers can list their advertising services with details such as type, pricing, and availability. Business owners can browse services using search, filters, or category-based options to find the most suitable advertising solution.

- Business owners can book services, and the platform tracks the booking status from pending, confirmed, active, completed, to cancelled. This ensures clear communication and reduces disputes.
- The system records all payments and updates their status as pending, completed, failed, or refunded. This helps both business owners and service providers track transactions efficiently.
- Business owners can create, edit, and monitor campaigns across multiple channels (e.g., billboards, vehicle branding, social media). The platform tracks campaign status and performance in real time.
- After completing a booking, business owners can rate and review service providers. This promotes accountability and helps other businesses choose reliable services.
- Administrators can manage all users, service listings, campaigns, and bookings. The dashboard provides tools for platform oversight and ensures smooth operation of the system.

5 Data Collection

The data for the ADLink system is collected based on direct experience, observation, and industry knowledge. The following methods are used:

- **Digital Marketing Experience:** The team are providing digital marketing services to multiple e-commerce businesses. This hands-on experience has helped gather practical data on service requirements, campaign management, pricing, and client preferences.
- **Experience-Based Data Collection :** Information about advertising trends, influencer services, and online promotions has been gathered through nearly three years of direct experience in marketing and scaling brands, by observing campaigns and marketing activities across various social media platforms.
- **Traditional Marketing Insights:** For offline advertising such as billboards and vehicle branding, data has been gathered by observing marketing practices in local surroundings and applying basic marketing principles. This includes understanding which businesses target offline audiences and how traditional media is used effectively.
- **Expert Knowledge Application:** General marketing knowledge and industry best practices are applied to structure the system, categorize services, and define campaign workflows for both online and offline advertising.

6 Technology Stack

The following technologies form the complete stack for the ADLink platform:

Layer	Technology / Purpose
Database (DBMS)	SQLite via SQL.js 1.10.3 — in-browser relational database engine with full SQL support including JOINS, triggers, and aggregations
Frontend Framework	React 18 (Hooks) — component-based UI using useState, useEffect, useRef, and useCallback
Styling	CSS custom dark theme with responsive grid layouts
Database Access Layer	Custom <code>q()</code> / <code>run()</code> / <code>insert()</code> helpers — parameterized queries to prevent SQL injection
Build Tool	Vite + Babel (JSX transform) — fast development server with hot module replacement
Fonts	Google Fonts CDN — Syne (headings) + DM Sans (body text)
Version Control	Git / GitHub — source code management and team collaboration
Production DB (future)	Firebase Firestore — cloud-hosted NoSQL DB for deployment as per SRS SW-1
Mobile (future)	Flutter — cross-platform mobile app for iOS and Android (SRS SW-3)

7 System Functionality

The ADLink Multi-Channel Advertising Management Platform provides the following key functionalities to its users: :

1. **User Registration and Role-Based Access:** All users—including business owners, service providers, asset owners, and administrators—can register and log in securely. The system ensures role-based access so that each user can only access the features relevant to their role.
2. **Service Listing and Marketplace:** Service providers can add advertising services with details such as type, pricing, and availability. Business owners can search, filter, and browse services by category to find the most suitable advertising solutions.
3. **Booking Management:** Business owners can book services, and the platform tracks the booking status through the full lifecycle: pending, confirmed, active, completed, or cancelled. This ensures clear communication and reduces disputes.
4. **Payment Recording, Status Management, and Withdrawal:** The system records all payments made through a secure payment gateway and tracks their status, including pending, completed, and failed, providing transparency for both business owners and service providers. Service providers can also withdraw their earned payments directly from the platform to their linked bank accounts, ensuring smooth and timely fund access.
5. **Multi-Channel Campaign Management:** Business owners can create, edit, and monitor campaigns across multiple channels such as billboards, vehicle branding, and

social media. Campaign status and performance are tracked in real time.

6. **Ratings and Reviews:** After completing a booking, business owners can provide ratings and reviews for service providers. This promotes accountability and helps other businesses make informed decisions when selecting services.
7. **Admin Dashboard and Oversight:** Administrators can manage users, service listings, campaigns, and bookings. The admin dashboard provides tools for monitoring platform activity, ensuring smooth operation, and resolving issues efficiently.
8. **User Profile Management:** All users can update their profile information including name, contact details, city, bio, and password.

8 User Classes

ADLink defines FOUR distinct user roles, each with specific access privileges and responsibilities within the platform:

Role	Access Level	Key Responsibilities
Admin	Full Platform Control	Manage all users, service providers, campaigns, bookings, payments, and reviews. Oversee the entire platform to ensure smooth operations.
Business Owner	Client / Buyer	Browse the advertising marketplace, book services, manage campaigns across multiple channels, make payments, and submit ratings and reviews for completed bookings.
Advertiser	Service Provider	List digital marketing services, manage incoming bookings, update service availability, and track payments efficiently.
Asset Owner	Service Provider	List physical advertising assets (billboards, vehicles, or other media), manage bookings, update availability, and track active and upcoming reservations.

9 Group Work Distribution

For our group project, we organized the work so that everyone stays involved throughout the whole process, from the proposal to the final exhibition. Each person is responsible for leading certain parts of the project to keep things clear and organized. However, our main goal is that everyone understands the entire project, not just the part they worked on.

To make sure of this, we meet every week to discuss our progress. The person who worked on a task explains their ideas, the decisions they made, and how they completed it. This helps the rest of the team understand the work. So even if one person wrote the code or created a diagram, everyone in the group can still explain it clearly.

Member 1

Member 1 is taking ownership of the database's structural and theoretical foundation. That means drafting the Problem Statement and Objectives for the proposal, then later leading the Normalization process to make sure our data properly meets 3NF requirements. During implementation, they will also be the one tackling the more complex SQL — Joins and Aggregations — and then breaking those down for the rest of the group so everyone fully understands them.

Member 2

Member 2 is focused on how the system actually works and feels. Early on, they will define the System Functionality and User Roles, then move into writing the DDL scripts to build out our schema in the DBMS. When we get to the application layer, Member 2 will lead the GUI development — building the interface that ties everything together on the front end.

Member 3

Member 3 is handling the technical environment and keeping everything documented and organized. They are researching our Technology Stack and managing the GitHub repository — making sure our commit history is clean, the README is clear, and everything is well maintained. Member 3 is also leading data collection and the database connectivity logic, and will be responsible for making sure everyone's contributions are properly merged and that our final report is polished and properly cited before we submit.

10 Conclusion

The ADLink Multi-Channel Advertising Management Platform offers a centralized, efficient, and transparent solution for managing both online and offline advertising services. By integrating business owners, service providers, and asset owners into a single platform, ADLink addresses the challenges of manual processes, fragmented service management, and complex campaign tracking.

The platform enables businesses—including small e-commerce stores and startups—to easily discover high-quality advertising services, manage multi-channel campaigns, and make secure bookings, while service providers can reach clients efficiently and maintain their offerings effectively. With features such as role-based access, booking lifecycle management, payment tracking, and ratings and reviews, ADLink ensures smooth operations and reduces disputes.

Overall, the system allows businesses to focus on improving products and scaling operations, while providing service providers with better visibility and client engagement. By streamlining advertising management and consolidating all services in one platform, ADLink promises increased efficiency, transparency, and growth opportunities for all stakeholders.

11 References

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